GG-P-133b
May 24, 1966

SUPERSEDING
Int. Fed. Spec. GG-P-00133a (GSA-FSS)
June 22, 1964 and
Fed. Spec. GG-P-133
July 20, 1955

FEDERAL SPECIFICATION

PEN, RULING, DRAFTSMAN'S: RAILROAD, SWIVEL BLADE, 5-INCH

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of swivel-blade railroad ruling pen used in drafting work for drawing double-line railroad and road symbols.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-585—Copper-Nickel-Zinc Alloy Plate, Sheet, Strip, and Bar (Copper Alloy Numbers 735, 745, 752, 762, 766 and 770).
QQ-C-586—Copper-Nickel-Zinc Alloy; Rod, Shapes, and Flat Products with Finished Edges (Flat Wire, Strip, and Bar).
QQ-S-766—Steel Plates, Sheets, and Strip—Corrosion Resisting.
TT-I-528—Ink, Drawing, Waterproof, Black.
UU-P-553—Paper, Wrapping, Tissue.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Setup.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-B-121—Barrier Material, Grease-Proof, Water Proof Flexible.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The pen shall consist of two pairs of pen blades with a tubular handle constructed so that the pairs of pen blades shall turn freely or remain fixed on the handle as desired. The pen blades shall be curved so that the points are eccentric to the center line of the handle, similar to the design of a contour pen in order that the pen blades will readily follow sharp curves. The overall length of the pen, including the handle, shall be 5 ± 1/4 inch. The pen shall be designed to provide lightness, good balance, and ease in handling.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of commercial quality, suitable for this type of equipment.

3.2.1 *Copper-nickel-zinc alloy.* Copper-nickel-zinc alloy shall conform to QQ-C-585, composition No. 1, rolled temper, or QQ-C-586, composition No. 1, cold drawn. Temper shall be the best suited for the use.

3.2.2 *Corrosion-resisting steel.* Corrosion-resisting steel shall conform to QQ-S-766, class 301 or 430, condition and temper best suited for the use.

3.3 Pen blades. The two pairs of pen blades shall be of the spring-blade type, and shall be fabricated of corrosion-resisting steel or spring steel coated to prevent corrosion with or without tips of tungsten high-speed steel. The pen nibs shall have a Rockwell hardness of C55 ± 5. The shank end of the pen blades shall be rigidly attached to one end of the handle rod specified in 3.4. Each pair of pen blades shall be bowed to bring the two pairs of blades parallel. Each pair of pen blades shall be provided with a thumbscrew for adjusting the distance between the blade points. The spacing between the two pairs of blades shall be adjustable from 1/32 to 3/8 inch. The pen blades shall be accurately set, evenly ground, and properly sharpened. The pen blades shall permit drawing clean, unbroken and uniform parallel lines, without cutting into the drawing medium. The width of the individual lines shall be adjustable from 0.003 to 0.015 inch.

3.4 Handle rod. The handle rod shall be made of corrosion-resisting steel of a size suitable for the purpose. The rod shall extend the entire length of the handle tube to permit easy turning and following of the smallest curve. The rod shall be threaded at the free end to provide for a clamp nut.

3.5 Handle tube. The handle tube shall be copper-nickel-zinc alloy, corrosion-resisting steel, ebony, or aluminum, as specified (see 6.1). If aluminum is used, it shall be anodized. The inside diameter of the tube shall be of such size as to permit the handle rod to rotate freely. The ends of the tube shall be fitted with suitable tips. The handle tube and the handle rod shall remain stationary with respect to each other when the clamp nut is tightened.

3.6 Adjusting nut and thumbscrews. The adjusting nut at the top of the handle and thumbscrew heads shall be made of copper-nickel-zinc alloy or corrosion-resisting steel. Screw studs shall be made of corrosion-resisting steel. The threads shall be such size and pitch to produce smooth action and withstand ordinary conditions of use.

3.7 Identification marking. Each pen shall have the manufacturer's name or trademark stamped or otherwise permanently marked on the handle.

3.8 Finish. The pens shall be finished in accordance with the manufacturer's standard practice for this type of equipment.

3.9 Workmanship. The pen, ruling, draftsman's shall have a uniform, unmarred fin-

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ish. Each component shall be constructed to facilitate an accurate fit of mating parts. Internal and external parts shall be free from residues of fabrication, and otherwise conform to the manufacturer's standard practice for items of this type.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.2.1 *Examination.* Samples, selected in accordance with 4.2, shall be examined as follows. Lot acceptance or rejection shall be based on acceptable quality level (AQL) of 2.5 percent defective for major defects and an AQL of 6.5 percent defective for minor defects.

Major:
- (a) Material not as specified.
- (b) Pen blades not as specified (3.3).
- (c) Handle rod not as specified (3.4).
- (d) Handle tube not as specified (3.5).
- (e) Workmanship not as specified (3.9).

Minor:
- (a) Finish not as specified (3.8).
- (b) Identification marking not as specified (3.7).

4.2.2 *Sampling.* Sampling for test shall be in accordance with MIL-STD-105.

4.3 Test.

4.3.1 *Adjusting nut and thumbscrews.* The adjusting nut shall be manually turned to determine freedom of movement and adjustment of the blades. The thumbscrews shall be manually tightened and loosened to determine the adjustment of the open nibs.

4.3.2 *Operational test.* The assembled pen, after the test in 4.3.1, shall be filled with drawing ink conforming to TT-I-528 and tested by drawing lines of various widths throughout spacing ranges of 1/32 to 3/8 inch on drawing paper or tracing cloth. Prior to the ink test, the drawing medium shall be cleaned with art gum or commercial cleaner. After the ink has dried, the lines and the spacing shall be measured to determine conformance with 3.3.

4.3.3 *Hardness test.* The hardness of the metal used shall be tested in accordance with Fed. Test Method Std. No. 151.

4.3.4 *Acceptance tests.* Samples, selected in accordance with 4.2.2, shall be tested as specified in 4.3.1, 4.3.2, and 4.3.3. Lot acceptance or rejection shall be based on an AQL of 2.5 percent defective. Samples failing to pass any of the tests shall be considered defective.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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TABLE I. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A.*

5.1.1.1 *Cleaning and drying.* The pens shall be cleaned by process C-1 and dried by procedure D-4 in accordance with MIL-P-116.

5.1.1.2 *Preservative application.* The pen blades shall be coated with type P-9 preservative in accordance with MIL-P-116.

5.1.1.3 *Unit packaging.* Each pen shall be wrapped with material conforming to MIL-B-121, grade A, class 2 and packaged method IA-15 in accordance with MIL-P-116.

5.1.1.4 *Intermediate packaging.* Unless otherwise specified (see 6.1), 12 unit packages of pens shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, W5c. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 *Level B.*

5.1.2.1 *Unit packaging.* Each pen shall be wrapped with material conforming to UU-P-553, type II and packaged in a close-fitting box conforming to PPP-B-566, style IX; PPP-B-665, style C; or PPP-B-676, type I.

5.1.2.2 *Intermediate packaging.* Twelve pens shall be packaged in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specifications.

5.1.3 *Level C.* The pens shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Intermediate packages of pens, in quantities as specified (see 6.1), shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed and strapped in accordance with the appendix to the box specification.

5.2.2 *Level B.* Intermediate packages of pens, in quantities as specified (see 6.1), shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 *Level C.* The pens shall be packed to insure carrier acceptance and safe delivery at destination in containers at the lowest rates complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* In addition to markings required by the contract or order,

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the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and procurement documents should include the following information:

(*a*) Title, symbol, and date of this specification.

(*b*) Material as specified (see 3.5). Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

(*c*) Intermediate package quantities required in shipping container (see 5.2.1 and 5.2.2).

(*d*) Selection of applicable level of preservation, packaging, and level B preservation and packaging is intended to provide economical but limited protection and should be specified only when the pens will be held in covered storage.

6.2 Federal specifications do not include all types, sizes, materials, finishes, and other descriptions on the items indicated by the title of the specifications, but are intended to cover those types, sizes, and so forth, which are most suitable for Federal Government requirements.

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